

Electrochemical-Thermal Modeling of a Gr–Si Composite Anode Cell

Joshua

February 6, 2026

Abstract

This document summarizes the full mathematical formulation of the volume-coupled Thevenin ECM and lumped thermal model used to simulate the electrical and thermal behavior of a graphite–silicon (Gr–Si) composite anode lithium-ion cell. The model includes SoC dynamics, OCV blending, volume-dependent resistance and capacitance, reversible and irreversible heat generation, nonlinear convection, and a two-branch RC network for dynamic overpotentials.

1 Introduction

This file documents the governing equations used in the MATLAB implementation of our coupled electrochemical and thermal model for a Gr–Si composite anode lithium-ion cell. The model includes:

- A two-branch Thevenin ECM with volume-dependent parameters,
- A blended OCV model for graphite and silicon,
- A lumped thermal model with reversible and irreversible heat,
- Nonlinear convection and Si-dependent thermal mass,
- A discharge–charge current profile.

2 State of Charge Dynamics

The anode state of charge evolves according to

$$\frac{dSoC}{dt} = -\frac{I(t)}{Q_a}, \quad (1)$$

where $SoC(t)$ is the anode SoC, $I(t)$ is the applied current, and Q_a is the anode capacity. This was the founding assumption for generating our theoretical current at a given point in time. **Note:** Q_a was a presumed value based on A.I. generation of a GrSi-NMC 21700class cell.

3 Open-Circuit Voltage Model

The composite OCV is a weighted mixture of graphite and silicon:

$$U_{\text{OCV}}(z, w) = w U_{\text{OCV}_{\text{Si}}}(z) + (1 - w) U_{\text{OCV}_{\text{Gr}}}(z), \quad (2)$$

where w is the silicon weight fraction and z is the SoC. dU/dt (Entropic coefficient) if not currently being used and was just part of a previous implementation; so, it will not be expanded upon here at this time.

4 Entropic (Reversible) Heat

The reversible heat generation is

$$Q_{\text{rev}} = I(t) T(t) \frac{\partial U_{\text{OCV}}}{\partial T}. \quad (3)$$

The entropic coefficient is blended as

$$\frac{\partial U_{\text{OCV}}}{\partial T} = w \frac{\partial U_{\text{Si}}}{\partial T} + (1 - w) \frac{\partial U_{\text{Gr}}}{\partial T}. \quad (4)$$

5 Volume Expansion Model

Silicon expansion induces volumetric strain:

$$\varepsilon(t) = a_{\text{Si}} w z(t), \quad (5)$$

where a_{Si} is the maximum strain coefficient. This equation is subject to change after the main volume expansion model is implemented. **DO NOT FORGET TO UPDATE**

6 Volume-Coupled ECM Parameters

Resistances and capacitances scale with strain:

$$R_i(t) = R_{i,0} (1 + k_{R_i} \varepsilon(t)), \quad (6)$$

$$C_i(t) = C_{i,0} (1 + k_{C_i} \varepsilon(t)), \quad (7)$$

for $i = 1, 2$. The total ohmic resistance is

$$R_{\text{tot}}(t) = R_0(t) + R_1(t) + R_2(t). \quad (8)$$

7 RC Network Dynamics

Each RC branch follows

$$\frac{dU_{RC_i}}{dt} = -\frac{U_{RC_i}}{R_i(t)C_i(t)} + \frac{I(t)}{C_i(t)}, \quad i = 1, 2. \quad (9)$$

8 Terminal Voltage

The terminal voltage of the cell is

$$U_{\text{term}}(t) = U_{\text{OCV}}(z(t), w) - I(t) R_0(t) - U_{RC_1}(t) - U_{RC_2}(t). \quad (10)$$

9 Heat Generation

Irreversible heat:

$$Q_{\text{irr}} = I(t)^2 R_{\text{tot}}(t). \quad (11)$$

Reversible heat:

$$Q_{\text{rev}} = I(t) T(t) \frac{\partial U_{\text{OCV}}}{\partial T}. \quad (12)$$

10 Nonlinear Convection

The convection coefficient increases with temperature:

$$hA(T) = hA_0 [1 + \alpha (T - T_{\text{amb}})]. \quad (13)$$

where hA_0 is currently between 0.2 and 0.4 and is set as a constant in our options.

11 Thermal Mass

The effective heat capacity of the composite material is

$$c_p(w) = 800(1 - w) + 700w, \quad (14)$$

and the thermal mass is.

$$C_{\text{th}}(w) = m c_p(w). \quad (15)$$

The Gr was set with 800J/kgK , and Si with 700J/kgK . The cell mass is currently put at 300g.

12 Thermal ODE

The full lumped thermal model is

$$\frac{dT}{dt} = \frac{Q_{\text{irr}} + Q_{\text{rev}} - hA(T) (T - T_{\text{amb}})}{C_{\text{th}}(w)}. \quad (16)$$

13 Current Profile

The discharge-charge transition is modeled as

$$I(t) = \begin{cases} I_{\text{DCH}}, & t \leq t_{\text{mid}}, \\ I_{\text{CH}}, & t > t_{\text{mid}}. \end{cases} \quad (17)$$

This is subject to change with our distributed current model.

14 Figures

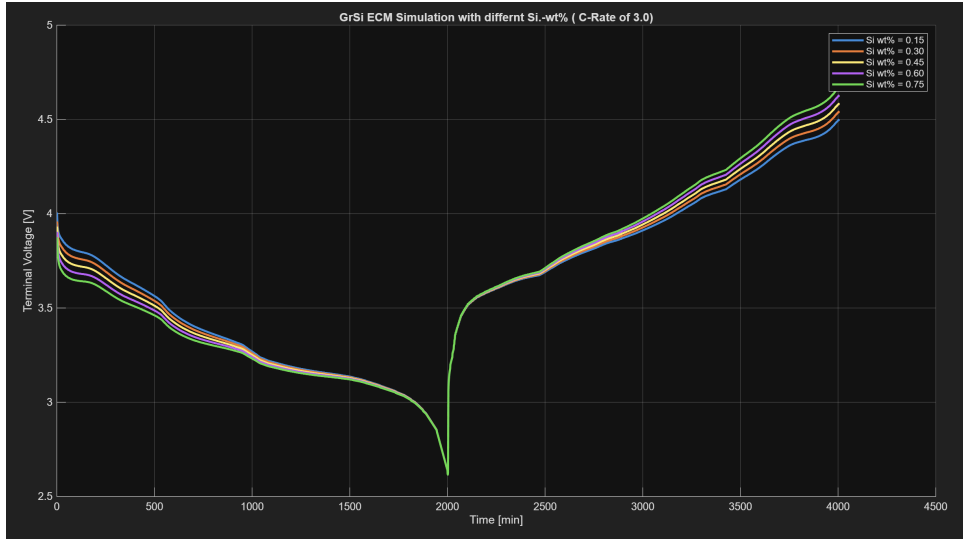


Figure 1: Terminal voltage for Volume Coupled Thevenin RC for multiple Si weight fractions at 3C.

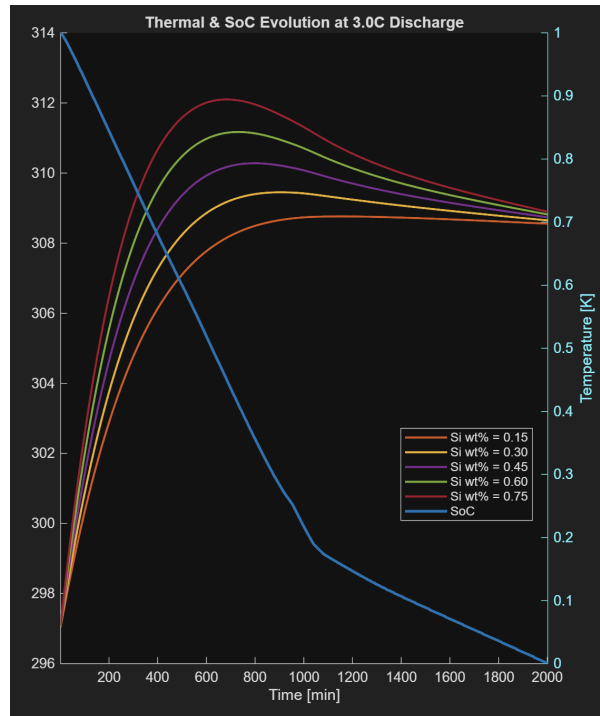


Figure 2: Thermal response for a 3C Discharge.

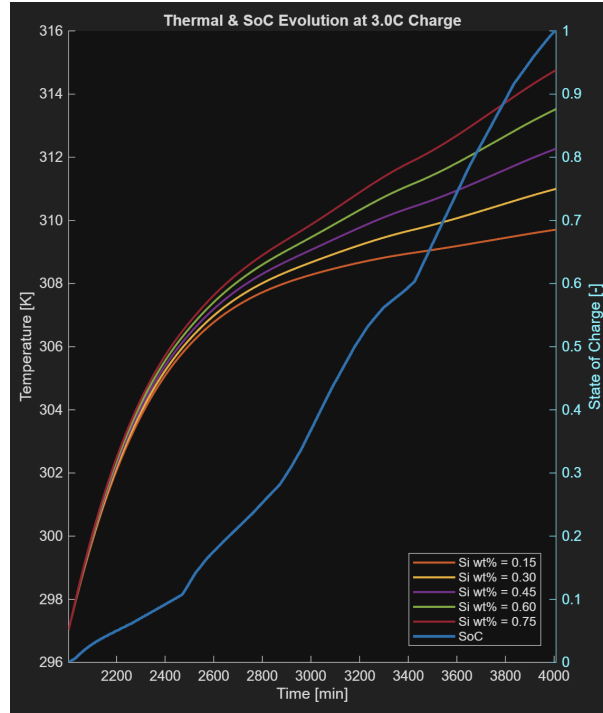


Figure 3: Thermal response for a 3C Charge.

15 Conclusion

This document is just meant as a placeholder. Will not be used on the report, but more so for updated abstracts and team awareness.